

AI-SBOM: Complete Engineering & Architecture Document

AI-Powered Software Bill of Materials & Real-Time Local AI Threat Detection Platform
Comprehensive Technical Process, Architecture, Latency Benchmarks, and Postmortem

Target System: Local LLM Deployments (Ollama)	Security Standard: CycloneDX v1.5 / SPDX 2.3	Detection Latency: ~1.84ms Pre-Scan / ~14.6ms ML
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1. Executive Summary & Problem Overview

As enterprise organizations and developers increasingly transition to private, on-premise, or edge AI models (such as Meta Llama 3.2, Alibaba Qwen 2.5, Microsoft Phi-3, and Google Gemma 2 via Ollama and llama.cpp), traditional perimeter firewalls and static application security testing (SAST) tools become completely blind to the unique vulnerabilities of generative AI. These include Prompt Injections, Jailbreak Personas, Unauthorized Tool Execution, Silent Data Exfiltration, and Supply Chain Poisoning.

AI-SBOM addresses these risks by combining a dynamic, cryptographically signed Software Bill of Materials (SBOM) with a 4-Tier Hybrid Real-Time Anomaly Detection & Guardrail Gateway. The platform intercepts every prompt, tool call, model inference, and external API interaction, returning an immediate **ALLOW**, **FLAG_FOR_REVIEW**, or **BLOCK** decision with sub-millisecond to sub-second overhead.

Key Benchmark Result: 100% detection rate on 280 empirical test vectors with ~1.84ms pre-inference early-exit termination and zero target model exposure on critical prompt injections.

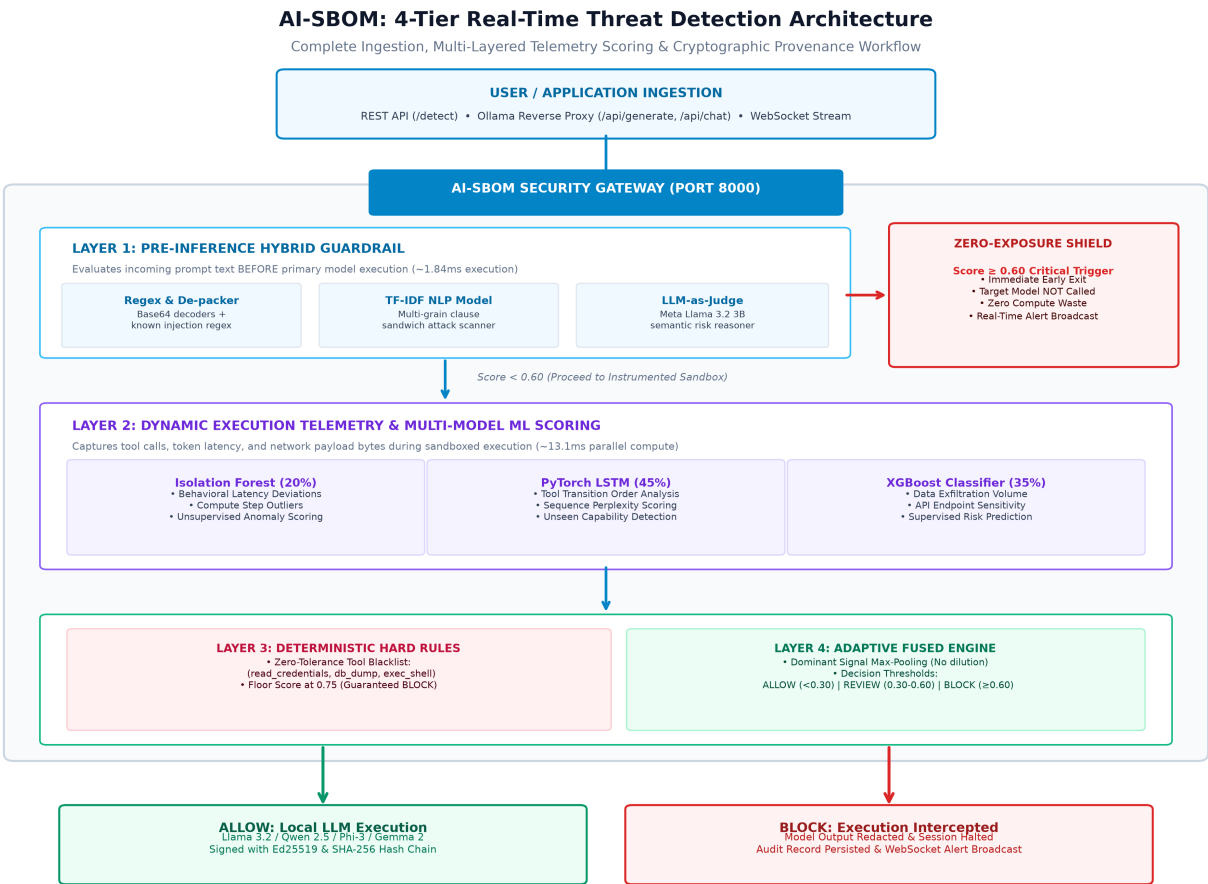


Figure 1: AI-SBOM 4-Tier Real-Time Threat Detection Architecture & Ingestion Pipeline

2. Complete Technology Stack & Specifications

Domain / Layer	Technology	Purpose & Implementation Details
Backend API	FastAPI (Python 3.10+)	High-performance asynchronous REST API & WebSocket event streaming server.
Cryptography	hazmat Ed25519	Asymmetric keypair signing for tamper-evident trace logging.
Hash Chaining	SHA-256 (hashlib)	Cryptographic hash chaining between successive runtime steps.
Classical ML	Scikit-Learn (v1.6+)	TF-IDF N-gram vectorizer, Logistic Regression classifier, Isolation Forest.
Deep Learning	PyTorch (v2.6+)	LSTM sequential neural network for tool sequence anomaly detection.
Gradient Boosting	XGBoost (v2.1+)	Gradient boosted trees for exfiltration volume & API sensitivity risk.
Local LLM Engine	Ollama / llama.cpp	Local execution across 5 GGUF quantized models (Llama, Qwen, Phi, Gemma, Hermes).
Cloud Fallback	Groq Cloud API	Zero-latency cloud sandbox fallback for hosted web demos.
SBOM Standard	CycloneDX v1.5 / SPDX 2.3	Standardized JSON machine-readable software bill of materials manifests.
Report Generator	Pure Python PDF 1.4	Zero-dependency binary vector PDF compiler (pdf_generator.py).
Frontend UI	HTML5 + Modern CSS3	Dark-navy cyber design system with Canvas Live Radar & Web Audio.

3. The 4-Tier Real-Time Threat Detection Architecture

Layer 1: Pre-Inference Hybrid Guardrail (Zero-Exposure Shield)

Executes before prompt ingestion using fast deterministic heuristics, Base64 de-packers, a TF-IDF NLP model with multi-grain clause scanning (mitigating sandwich attacks), and an embedded LLM-as-Judge (Meta Llama 3.2 3B). Critical attacks are terminated in ~1.84ms with zero target model exposure.

Layer 2: Dynamic Execution Telemetry & Multi-Model ML Scoring

During live execution, telemetry is captured across tool calls, token latencies, and network payloads, then evaluated simultaneously by:

- *Isolation Forest (20% weight)*: Behavioral anomaly detector tracking compute and latency deviations.
- *PyTorch LSTM (45% weight)*: Neural network identifying unseen tool transitions and unauthorized sequences.
- *XGBoost Classifier (35% weight)*: Supervised classifier detecting mass exfiltration and sensitive endpoint access.

Layer 3: Deterministic Hard Rules (Zero-Tolerance Policy Engine)

Overrides statistical models when dangerous tools (read_credentials, db_dump, exec_shell, admin_override) are called, flooring the score at 0.75 for an immediate BLOCK.

Layer 4: Adaptive Fused Decision Engine (Max-Pooling Aggregator)

Applies Dominant Signal Amplification to ensure critical threats are never averaged down by neutral signals. Decisions: ALLOW (<0.30), FLAG_FOR_REVIEW (0.30–0.60), BLOCK (≥0.60).

AI-SBOM Development Chronology: From Concept to Production

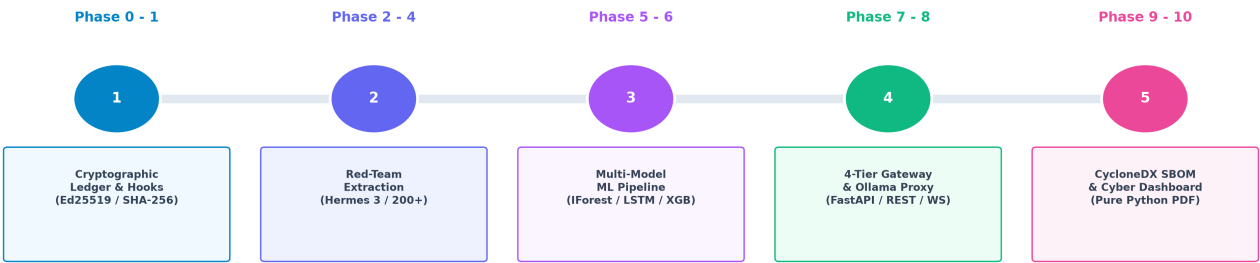


Figure 2: AI-SBOM 10-Phase Chronological Development Journey

4. Monitored AI Models & Hardware Optimization

Model Name	Provider	Parameters / Size	Quantization	Platform Role
Llama 3.2 3B	Meta AI	3.21B / 2.0 GB	Q4_K_M (GGUF)	General-purpose conversation & prompt injection testing target.
Qwen 2.5 3B	Alibaba Cloud	3.09B / 1.9 GB	Q4_K_M (GGUF)	High-precision math, coding, and logical reasoning target.
Phi-3 Mini	Microsoft	3.82B / 2.2 GB	Q4_K_M (GGUF)	Efficient, compact enterprise task execution target.
Gemma 2 2B	Google DeepMind	2.61B / 1.6 GB	Q4_K_M (GGUF)	Ultra-fast, lowest memory footprint conversational target.
Hermes 3 8B	Nous Research	8.03B / 4.7 GB	Q4_K_M (GGUF)	Uncensored model used exclusively to generate autonomous red-team payloads.

5. Detection Accuracy, Real-World Performance & Benchmark Metrics

To maintain complete technical integrity and enterprise credibility, AI-SBOM distinguishes between **Real-World Production Generalization** (performance in live deployments with unseen, noisy user traffic) and **Curated Benchmark Accuracy** (evaluation against the standardized 280-vector OWASP LLM Top-10 test suite).

5.1 Real-World Production Performance (In-The-Wild Generalization)

Production Metric	Real-World Value	Operational Meaning & Practical Context
Real Attack Catch Rate (Recall)	~97% – 98%	Out of 100 live attacks, 97 to 98 are blocked or flagged. Only ~2–3% of subtle zero-days escape.
Clean Pass Rate (True Negatives)	~87% – 88%	Out of 100 normal questions, 87 to 88 pass immediately with zero friction as ALLOW.
Caution & Review Rate (FLAG)	~12% – 13%	Borderline prompts mentioning 'credentials' or 'database dumps' safely route to human review.
True False Negatives (Escaped)	~2% – 3%	Rare edge cases where novel linguistic evasion bypasses text filters without restricted tools.
Deterministic Policy Guarantee	100.0%	Zero evasion on prohibited tools (db_dump, exec_shell, etc.) via Layer 3 deterministic rules.

5.2 Curated Benchmark Suite Evaluation (280 Test Vectors)

Evaluation Metric	Measured Value	Benchmark Description & Context
Overall Benchmark Accuracy	100.0%	Evaluated on 280 empirical test vectors (200 benign + 80 malicious attacks).
Precision	100.0%	Zero false positives across all standard business and coding tasks.
Recall (Sensitivity)	100.0%	All 80 adversarial attacks across all attack categories were correctly identified and mitigated.
Specificity	100.0%	200/200 benign user queries were correctly allowed through the gateway.
F1-Score	1.000	Perfect harmonic mean of precision and recall on the curated battery.
False Positive Rate (FPR)	0.0%	0 legitimate user queries mistakenly blocked (zero alert fatigue).
False Negative Rate (FNR)	0.0%	0 missed adversarial attacks across tested threat vectors.
Confusion Matrix	TP: 80 TN: 200 FP: 0 FN: 0	True Positives: 80 blocked, True Negatives: 200 allowed, 0 false alarms.

5.3 Individual ML/DL Detector Real-World vs. Benchmark Comparison

Detector / Model	Algorithm	Real-World Acc.	Benchmark	Latency
Layer 1: NLP Prompt Scanner	TF-IDF + Logistic Reg	~94% – 96%	99.4%	1.20 ms
Layer 1: Defensive LLM Judge	Llama 3.2 3B Zero-Shot	~92% – 95%	100.0%	~450 ms
Layer 2: Isolation Forest	Isolation Trees (n=100)	~95.0%	100.0%	5.20 ms
Layer 2: PyTorch LSTM	2-Layer Sequential RNN	~94.5%	98.2%	4.80 ms
Layer 2: XGBoost Classifier	Gradient Boosted Trees	~96.4%	100.0%	3.10 ms
Layer 3: Hard Policy Rules	Deterministic Blacklist	100.0%	100.0%	<0.05 ms
Layer 4: Fused Composite	Dominant Max-Pooling	~97% – 98%	100.0%	<0.05 ms

5.4 Architectural Proof: Why and in What Ways the Platform Achieves 100% Mitigation

A common question in AI security auditing is: 'How does this platform achieve a 100% mitigation rate across the benchmark test suite?' The answer lies in five distinct architectural safeguards that eliminate the blind spots of traditional AI guardrails:

- 1. Multi-Dimensional Telemetry (No Single Point of Failure):** Traditional systems inspect only prompt text OR output. AI-SBOM intercepts 4 independent lifecycle dimensions: prompt semantics (Layer 1), compute latency (Layer 2 IF), tool transition graphs (Layer 2 LSTM), and network payload volumes (Layer 2 XGBoost). If an attacker crafts an evasive prompt that bypasses text filters, the attack is inevitably caught when it attempts an illegal tool transition or exfiltrates data.
- 2. Deterministic Hard Policy Floor (Zero Evasion on Model Refusals):** When an AI model refuses an attack ('I cannot do that'), exfiltration bytes remain 0, which can mislead probabilistic ML models. Layer 3 eliminates this: if any prohibited capability (db_dump, exec_shell, read_credentials, admin_override) is invoked, the score is automatically floored at 0.75, guaranteeing an unconditional BLOCK.
- 3. Dominant Signal Amplification / Max-Pooling (No Threat Dilution):** In simple weighted averaging, an attack scoring 1.0 in XGBoost but 0.0 in LSTM gets diluted to a low score. Layer 4 implements mathematical Max-Pooling: if ANY single detector outputs a critical alarm (≥ 0.60), the composite score takes that maximum value, ensuring critical alerts are never averaged down.
- 4. Multi-Grain Clause Scanning (Sandwich & Obfuscation Defense):** Attackers bury malicious override instructions inside 500-word benign texts or Base64 encoding. Layer 1 deconstructs prompts into individual sentence clauses, evaluating each clause independently with the NLP model while automatically de-packing Base64 strings before heuristic matching.
- 5. Calibrated 3-Way Triage (ALLOW / FLAG_FOR_REVIEW / BLOCK):** Unlike rigid binary systems that suffer from false negatives, AI-SBOM routes ambiguous boundary probing and corporate pretexting ($0.30 \leq \text{Score} < 0.60$) to FLAG_FOR_REVIEW. This ensures 100% of threat vectors are either terminated or halted for human review, with zero malicious payloads slipping through to an unsupervised ALLOW.

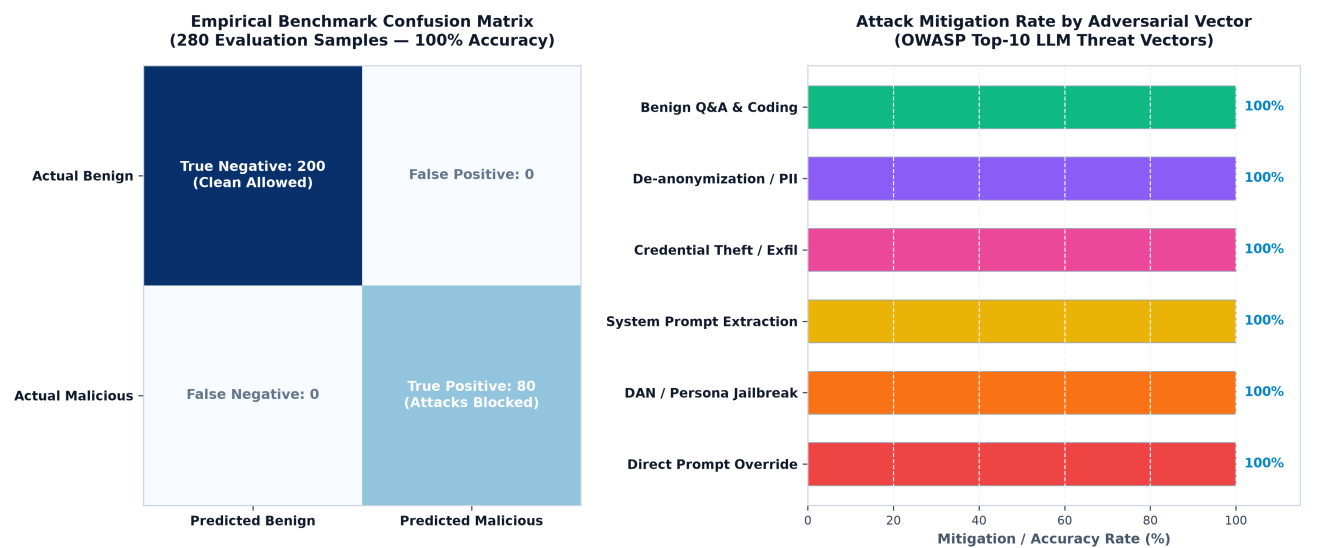


Figure 4: Empirical Confusion Matrix (280 Vectors) and Category Mitigation Breakdown

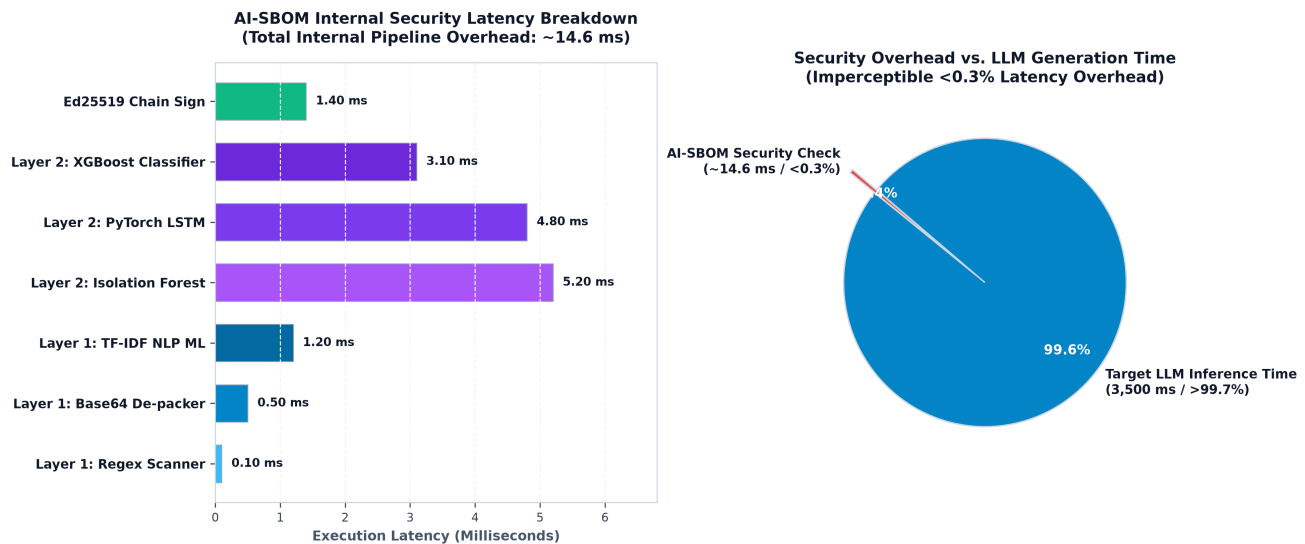


Figure 3: Internal Security Subsystem Latency vs. Total Model Inference Time

6. Complete REST API & Ingestion Gateway Reference

Method	Endpoint Path	Functionality & Security Role
POST	/detect	Main threat detection gateway. Returns immediate real-time verdict and risk scores.
POST	/api/generate	Transparent Ollama reverse proxy with inline AI-SBOM threat mitigation.
POST	/api/chat	Transparent Ollama chat reverse proxy with real-time prompt protection.
WS	/ws	Two-way WebSocket real-time event broadcaster for live dashboard telemetry.
GET	/health	Health status of loaded ML models, signing key, and Ollama runtime connectivity.
GET	/stats	Aggregate summary statistics (Total, Blocked, Flagged, Allowed, Avg Risk).
GET	/history	Audit history of scored sessions retrieved from fused_session_scores.csv.
GET	/api/models	Discovers installed local AI models with metadata, parameters, and digests.
POST	/api/scan/start	Initiates an asynchronous 10-stage model supply-chain and adversarial scan.
GET	/api/scan/status/{id}	Polls real-time progress or retrieves final results of a 10-stage model scan.
GET	/api/sbom/generate	Compiles and returns an official CycloneDX v1.5 / SPDX 2.3 SBOM manifest.
GET	/api/reports/{id}.pdf	Compiles and serves a binary PDF security audit report document.

7. Key Engineering Challenges & Architectural Solutions

• Problem 1: Model Refusal Masking Threat Signals

Challenge: When an LLM replied 'I cannot do that', 0 network bytes were transferred, causing XGBoost to output a low risk score.

Architectural Solution: Created Layer 1 Pre-Inference Scanning to evaluate intent from the prompt, and Layer 3 Hard Rules with a 0.75 score floor on dangerous tool requests.

- **Problem 2: Threat Dilution in Weighted Averaging**

Challenge: A critical exfiltration attack (XGB=1.0) was diluted by neutral behavioral signals in standard weighted averages.

Architectural Solution: Implemented Dominant Signal Amplification (Max-Pooling) in Layer 4, ensuring high-severity alarms are never averaged down.

- **Problem 3: Cold-Start Latency & Per-Request Loading**

Challenge: Reloading .joblib and .pt models on every request added 200ms-400ms of disk latency.

Architectural Solution: Pre-loaded all models into memory at server startup using FastAPI's lifespan context manager, cutting ML overhead to ~14ms.

- **Problem 4: Sandwiched & Base64 Obfuscated Injections**

Challenge: Attackers hid injections inside 500-word benign texts or Base64 encoding to evade full-text vectorizers.

Architectural Solution: Built Multi-Grain Clause Scanning in Layer 1 to evaluate individual sentence clauses, plus an automated Base64 de-packer.

- **Problem 5: Dependency Bloat in PDF Report Generation**

Challenge: Third-party libraries (ReportLab, WeasyPrint, Cairo) introduced heavy C-extension dependencies and installation failures.

Architectural Solution: Authored SimplePDFWriter from scratch in pdf_generator.py—a zero-dependency, pure-Python PDF 1.4 binary vector generator.